

years old to just 100,000 years old. Scientists didn't know how old the fossils are because they were found on the floor of a deep cave, not encased in rocks that could be dated using radioactive isotopes. In 2017, however, teeth from a newly discovered *H. naledi* skull were dated using radioactive isotopes and found to be 300,000 years old. Overall, fossil evidence suggests that *H. naledi* arose 1–2 million years ago (like other early members of our genus) and then persisted almost until the present.

About ten years before the discovery of *H. naledi*, researchers reported another astonishing find: skeletal remains of adult hominins dating from 100,000 to 60,000 years ago and representing a previously unknown species, *Homo floresiensis*. First discovered in a limestone cave on the Indonesian island of Flores, the individuals were much shorter and had a much smaller brain volume than *H. sapiens*—more similar, in fact, to an australopith. The researchers who discovered these fossils argue that certain features of the skeletons, such as the shape of the teeth and the thickness and proportions of the skull, suggest that *H. floresiensis* descended from the larger *H. erectus*. Not convinced, some researchers have argued that the fossils represent small *H. sapiens* individuals with a disorder such as Down syndrome or microcephaly (a condition in which a person has a deformed, miniature brain).

Over the past few years, most studies have supported the designation of *H. floresiensis* as a new hominin. One such study found that the wrist bones of the Flores fossils are similar in shape to those of nonhuman apes and early hominins, but different from those of Neanderthals and *H. sapiens*. These researchers concluded that the Flores fossils represent a species whose lineage branched off before the origin of the clade that includes Neanderthals and humans. A different study comparing the foot bones of the Flores fossils with those of other hominins also concluded that *H. floresiensis* arose before *H. sapiens*. Finally, in 2016, *H. floresiensis*-like fossils dating to 700,000 years ago were discovered at a second location on Flores—again indicating that the *H. floresiensis* lineage branched off long before the origin of *H. sapiens*. Open questions that may yet be answered by new discoveries include how *H. floresiensis* originated and whether it survived long enough to encounter *H. sapiens*, which was living in Indonesia by 50,000 years ago.

The rapid expansion of our species may have been spurred by changes in human cognition as *H. sapiens* evolved in Africa. Evidence of sophisticated thought in *H. sapiens* includes the recent discovery in South Africa of 77,000-year-old art—geometric markings made on pieces of ochre (**Figure 34.55**). Similarly, archaeologists working in southern and eastern Africa recently found 75,000-year-old ostrich eggs and snail shells with holes neatly drilled through them. By 30,000 years ago, humans were producing spectacular cave paintings.

While these developments can help us understand the spread of *H. sapiens*, it is not clear whether they played a

▼ **Figure 34.55 Art, a human hallmark.** The engravings on this 77,000-year-old piece of ochre, discovered in South Africa's Blombos Cave, are among the earliest signs of symbolic thought in humans.



role in the extinction of other hominins. Neanderthals, for example, also made complex tools and showed a capacity for symbolic thought. As a result, while some scientists have suggested that Neanderthals were driven to extinction by competition with *H. sapiens*, others question that idea.

Our discussion of humans brings this unit on biological diversity to an end. But keep in mind that our sequence of topics isn't meant to imply that life consists of a ladder leading from lowly microorganisms to lofty humanity. Biological diversity is the product of branching phylogeny, not ladderlike "progress." The fact that there are almost as many species of ray-finned fishes alive today as in all other vertebrate groups combined shows that our finned relatives are not outmoded underachievers that failed to leave the water. The tetrapods—amphibians, reptiles, and mammals—are derived from one lineage of lobe-finned vertebrates. As tetrapods diversified on land, fishes continued their branching evolution in the greatest portion of the biosphere's volume. Similarly, the ubiquity of diverse prokaryotes throughout the biosphere today is a reminder of the enduring ability of these relatively simple organisms to keep up with the times through adaptive evolution. Biology exalts life's diversity, past and present.

CONCEPT CHECK 34.7

1. Identify some characters that distinguish hominins from other apes.
2. Provide an example in which different features of organisms in the hominin evolutionary lineage evolved at different rates.
3. **WHAT IF?** Some genetic studies suggest that the most recent common ancestor of *Homo sapiens* that lived outside of Africa left Africa about 50,000 years ago. Compare this date with the dates of fossils given in the text. Can both the genetic results and the dates ascribed to the fossils be correct? Explain.

For suggested answers, see Appendix A.